

CASE STUDY

The Control Problem of a Global Toy Retailer

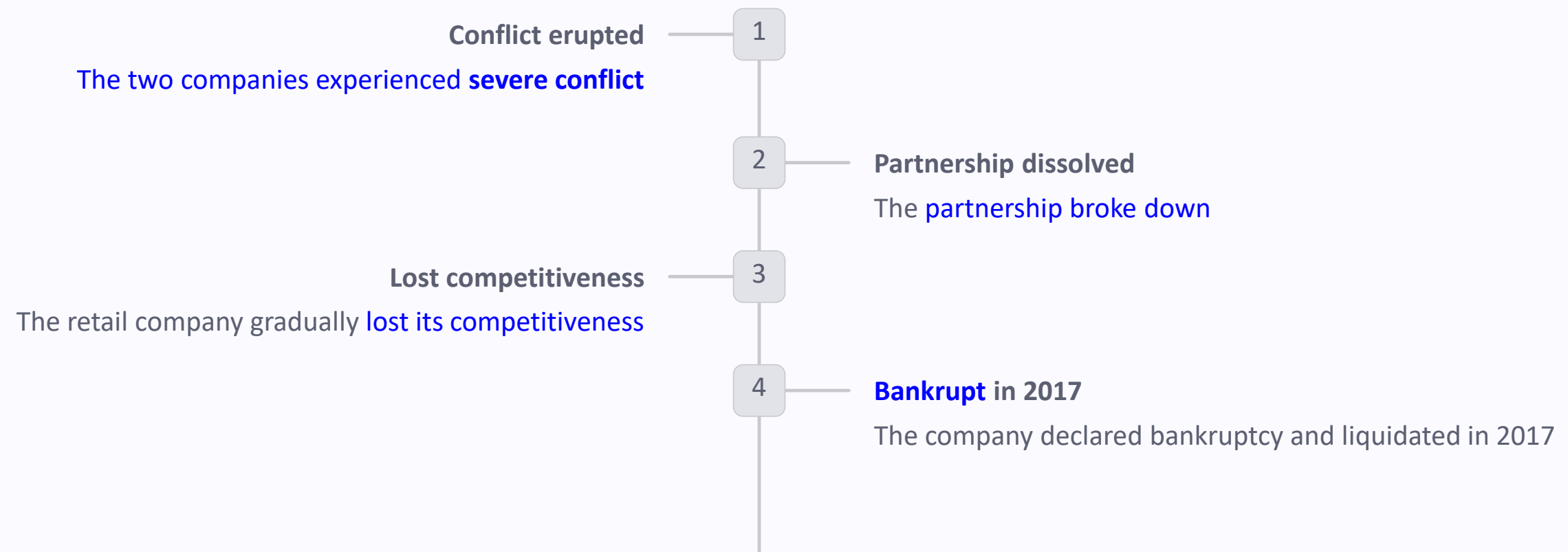
Today's case is a very famous retail example:

A company that was once the world's largest toy retailer made a decision in the early 2000s:

To partner with a fast-growing e-commerce company to manage its online store.

This collaboration seemed very successful at the time.

But a few years later:



The Question of Control in Digital Transformation

Many companies make a seemingly reasonable decision when going digital:

Seemingly Reasonable Choices

- Using mature platforms
- Letting professional companies handle systems
- Quickly entering the online market

In the short term, this offers many benefits:

- Reduced technical risk
- Lower costs
- Faster time-to-market

But the problem is:

If your core channels are built on someone else's platform, how much control do you really have?

For example, these questions are critical in the digital age:

Who controls customer data?

Who decides product selection?

Who sets platform rules?

Today we will discuss three questions

- 1 Why did this company make this cooperation decision in the first place?
- 2 What were the short-term and long-term impacts of this decision?
- 3 When undergoing digital transformation, how should enterprises build their digital architecture?

Before discussing the case, today we will start our lecture with the following three topics:

Re-architecting the Firm

How Organizational Structure
Affect System Design

Value Stream Integration

WEEK 3

Re-architecting the Firm

How Jeff Bezos's legendary **2002 memo transformed Amazon**—and redefined the operating architecture of the modern enterprise.

When Amazon Hit a Wall — What Bezos Did Next

Core Principles

1. Service Interfaces
2. Communication through interfaces only
3. No backdoors
4. Service interface calls Only
5. Externalizable Design

* **Design internal systems as if they were future products.** In fact, AWS was not invented later; Bezos had already embedded the architectural foundations of AWS within Amazon's internal systems.

FROM: Jeff Bezos
TO: All Development
SUBJECT: Bezos Mandate

All teams will henceforth expose their data and functionality through service interfaces. Teams must communicate with each other through these interfaces.

There will be no other form of inter-process communication allowed: no direct linking, no direct reads of another team's data store, no shared-memory model, no back-doors whatsoever. The only communication allowed is via service interface calls over the network. It doesn't matter what technology they use.

All service interfaces, without exception, must be designed from the ground up to be externalizable. That is to say, the team must plan and design to be able to expose the interface to developers in the outside world. No exceptions.

Anyone who doesn't do this will be fired. Thank you; have a nice day!

Jeff Bezos

When Amazon Hit a Wall —What Bezos Did Next

In 2002, Amazon's systems were starting to have serious problems. The company **had grown too fast** — selling books, office supplies, electronics, and clothing — and these products were sold across a **patchwork of systems** stitched together mainly by a **shared homepage**. There was **no consistent technology architecture, no unified data structure, and no single view of the customer**.

*A combination of various, often mismatched, elements joined together to form a whole.



The memo Bezos wrote that year became one of the most important documents in corporate digital transformation. It pointed to a deeper truth: the problem wasn't just growth — it **was how Amazon was built**.

What Makes a True Digital Enterprise?

Digital enterprises don't just use the internet or adopt mobile technology. What sets a true digital enterprise apart is that it is **architected differently** — built on a fundamentally new business and operational foundation.

The Old Mindset

Take existing processes and make them digital. Technology is a tool that supports the business.

The New Mindset

Every process is **essentially digital** from the beginning. **Software, data, and AI are the core of how the business runs — not an add-on.**

Unlike **traditional labor- and asset-intensive models**, digital enterprises use **data pipelines and algorithms** to create value — generating increasing returns at scale, scope, and speed. **Even the most advanced AI systems won't deliver valuable results unless the entire operating model is designed to support them.** Bezos understood this early: **Amazon's future depended on rebuilding its operating architecture** — the way its business processes were connected and structured — from the inside out.

- ❏ **The real challenge wasn't 'digitalizing' old processes. It was building something fundamentally different from the beginning — a new kind of company for the 21st century.**

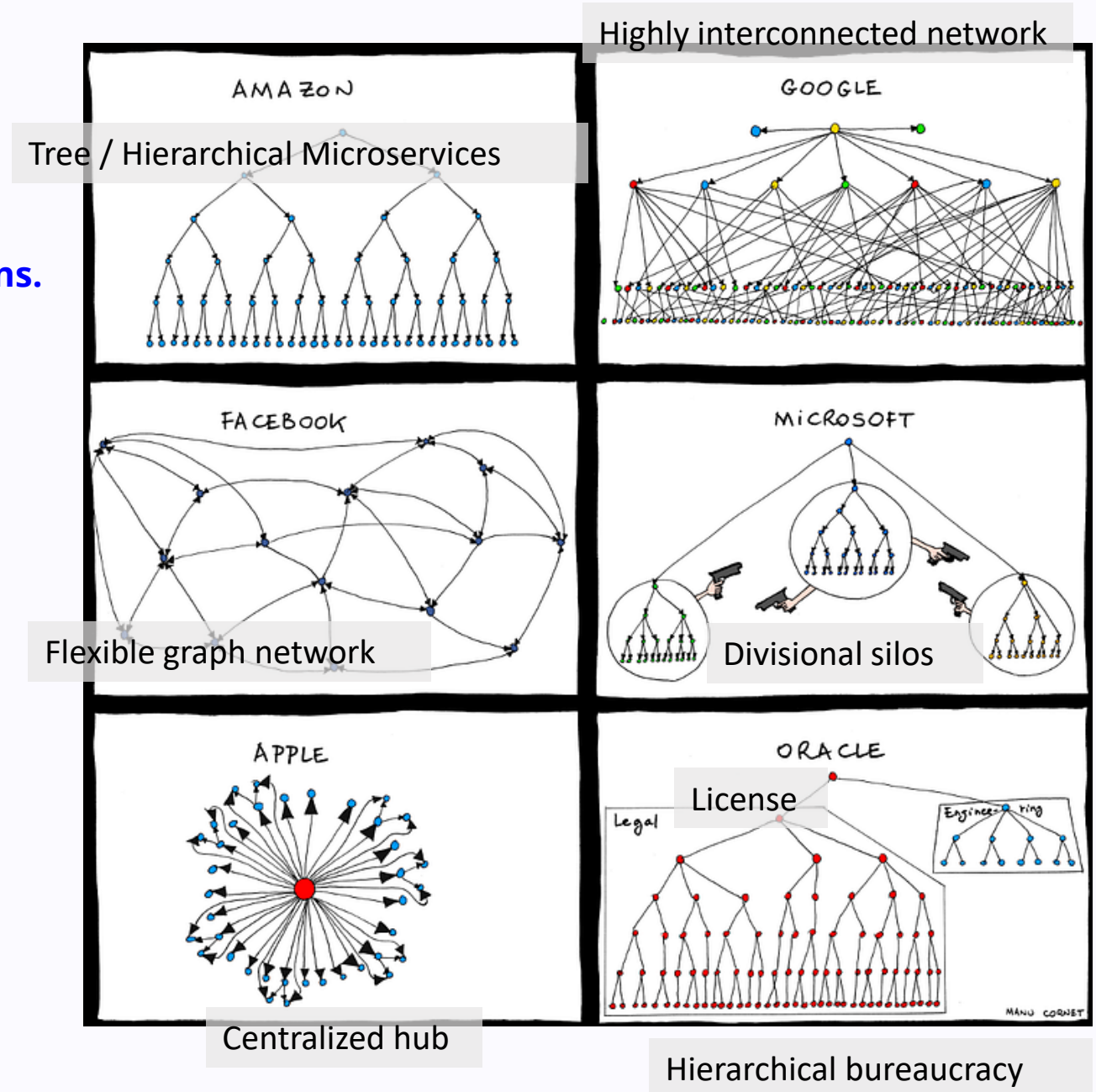
The Mirroring Hypothesis

In 1967, computer scientist **Melvin Conway** noted that **an organization is constrained to design systems that reflect its own communication patterns**. This is now known as **Conway's Law**.

The **mirroring hypothesis** states that **organizational ties within a project or firm tend to reflect the technical dependencies between tasks**. Beyond individual design tasks, **system architecture often mirrors the structure of the organizations that build the systems**.

This is why software projects often organize teams around features rather than functions, so that people responsible for closely related tasks can work together.

* Organizational ties: who needs to work with whom inside an organization.



When Strengths Become Constraints

As organizations perform similar tasks repeatedly, **they develop productive routines embedded in technology, processes**, and incentives. These patterns create very efficient routines, but those routines can **build inertia that stops the response to change**.



IBM: Mainframe to PC

IBM's failure to transition from mainframe to PC architecture opened the door for new competitors.



Microsoft & Smartphones

Microsoft's failure to rearchitect PCs into smartphones ceded mobile dominance to Apple and Google.

Henderson and Clark (1990) argued that **architectural innovations**—those **requiring changes between technological components—are a particular danger for established firms**. This concept informs Christensen's disruption theory: it is the **architectural inertia established by links with existing customers that prevents effective response to disruptive change**.

Breaking Architectural Inertia

In writing such a clear and provocative memo, Bezos was trying to break architectural inertia and change the architecture not only of Amazon's technology but also of Amazon's organization.

Bezos was determined to transform Amazon's operating architecture and build the foundations for a **software-, data-, and AI-driven firm**. To understand why this was so revolutionary, we must first look at the historical roots of operating models—and see why traditional architectures are as entrenched as they are.

The Origins of Siloed Operations

Long before information technology, **firms evolved into siloed operating architectures of specialized, largely autonomous units**. Dating back to the Italian Renaissance, operating models managed complexity by breaking organizations into smaller units, each given independence to minimize the load on excruciatingly slow communication lines.

15th Century — Prato, Italy

Wool and textile trades distributed operations across specialized production, distribution, banking, and insurance facilities—a loosely connected ensemble linked by family ties and joint ownership.

By 1670 — Global Powerhouse

Nearly 200 ships, 50,000+ employees, and a private army. Dominated global trade in spices, silk, cotton, porcelain, and textiles through a complex **siloed** operating model.

1

2

3

1602 — Dutch East India Co.

Possibly the first modern corporation. Consolidated seven rival trading companies, achieving economies of scale while evolving into a multi-unit, geographically distributed structure.

*** In this scenario, autonomy means that each department or group manages itself independently within its “own scope” , leading to a lack of collaboration and creating isolated "silos" within the organization.**

The Rise of Mass Production

The **Industrial Revolution transformed production techniques**. **Mass production meant each worker focused on a single component or stage**, benefiting from specialization and repetition to increase scale and speed of learning.

1 Ford Motor Company (1903)

Founded with \$28,000 from twelve investors. Ford's vision: make automobile transportation affordable and accessible to the average person.

2 The Model T (1908)

The "Tin Lizzie"—explicitly designed for mass production. Generally considered the first automobile within reach of most American consumers.

3 Moving Assembly Line (1913)

With Frederick Taylor's help, Ford cut Model T assembly time by **a factor of 10**. Prices dropped by more than half. By 1918, half of all cars in America were Model Ts.

Ford became the largest US manufacturer by deploying extraordinary levels of standardization and specialization—breaking functional specialties down to the smallest, most standardized human tasks.

The Double-Edged Legacy of Twentieth-Century Operating Models

Scale Economies & the Death of Craft

The economies of scale that powered industrial growth nearly eliminated traditional craftsmanship across a wide range of manufacturing and service industries — from weapons and textiles to agriculture and insurance — and eventually spread into the broader service sector. **They achieved great success, but also reached clear limits as things became too complex for the organization to handle in this way.**

Enterprise IT: More of the Same

Even as we entered the **21st century, corporate IT deployments** did not fundamentally change the trajectory of operating model evolution. Across successive waves of adoption — mainframes in the 1960s–70s, client-server in the 1980s, and early internet systems in the 1990s — IT systems like Oracle Finance and SAP PLM improved many traditional processes, but. **largely mirrored these siloed, specialized structures already in place.**

- ❏ **Key Insight:** These IT systems **essentially digitized existing departmental silos rather than breaking them down — processes remained fragmented,** and cross-functional collaboration was not meaningfully improved.

Traditional Operating Limits in Modern Tech Processes

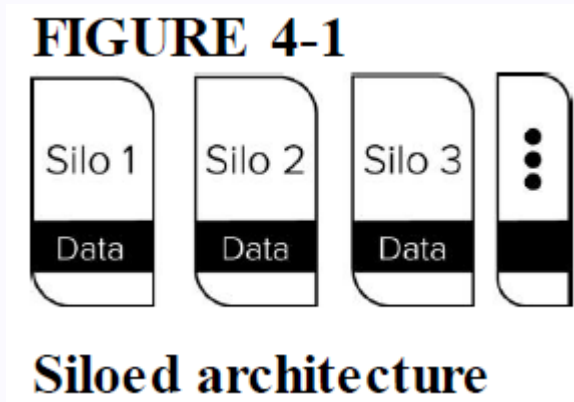


Figure 4-1: Siloed architecture. In company after company, processes, software applications, and **data remain embedded in largely autonomous, siloed organizational units**. Large firms often use thousands of enterprise applications with scattered databases and incompatible legacy technology. Integrating data across silos without rearchitecting the entire system is a long, horrifically complicated, unreliable process.

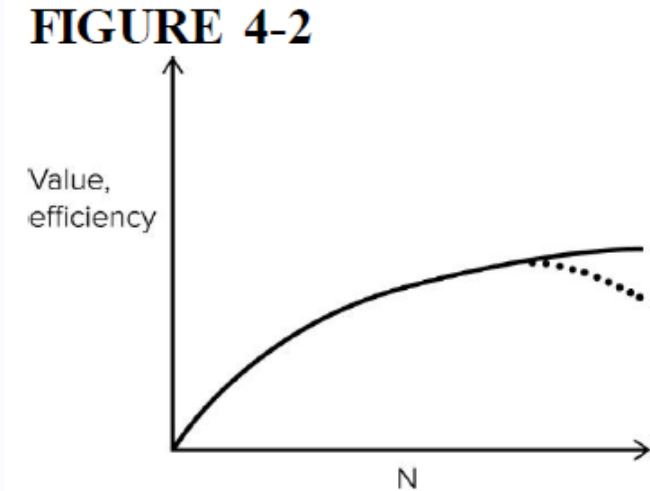


Figure 4-2: Diminishing returns. A traditional organization's ability to deliver value faces a curve of diminishing returns. **N represents variables such as the number of users or complementors** on a platform. Despite generations of management improvements and enterprise IT deployment, **operating model complexity constrains the value traditional firms can deliver.**

From Silos to Agile: How Bezos Rebuilt Amazon

Before Bezos wrote his famous memo, Amazon looked like a traditional company — its organization, data, and technology **had evolved into silos, with disconnected units and messy connections built only to fight fires.**

Bezos recognized that **technology should lead, and organization should follow** — **not only letting technology adapts to traditional organization framework.**

→ He **rebuilt Amazon's tech platform first, then restructured teams around it** to enable **agile collaboration.**

Two Ways to Organize Agile Teams

Cross-Functional Project Teams

Specialists are pulled from various departments and dedicated full-time to a specific project. Members collaborate long-term until the project concludes, enabling high flexibility, fast response, and smooth coordination.

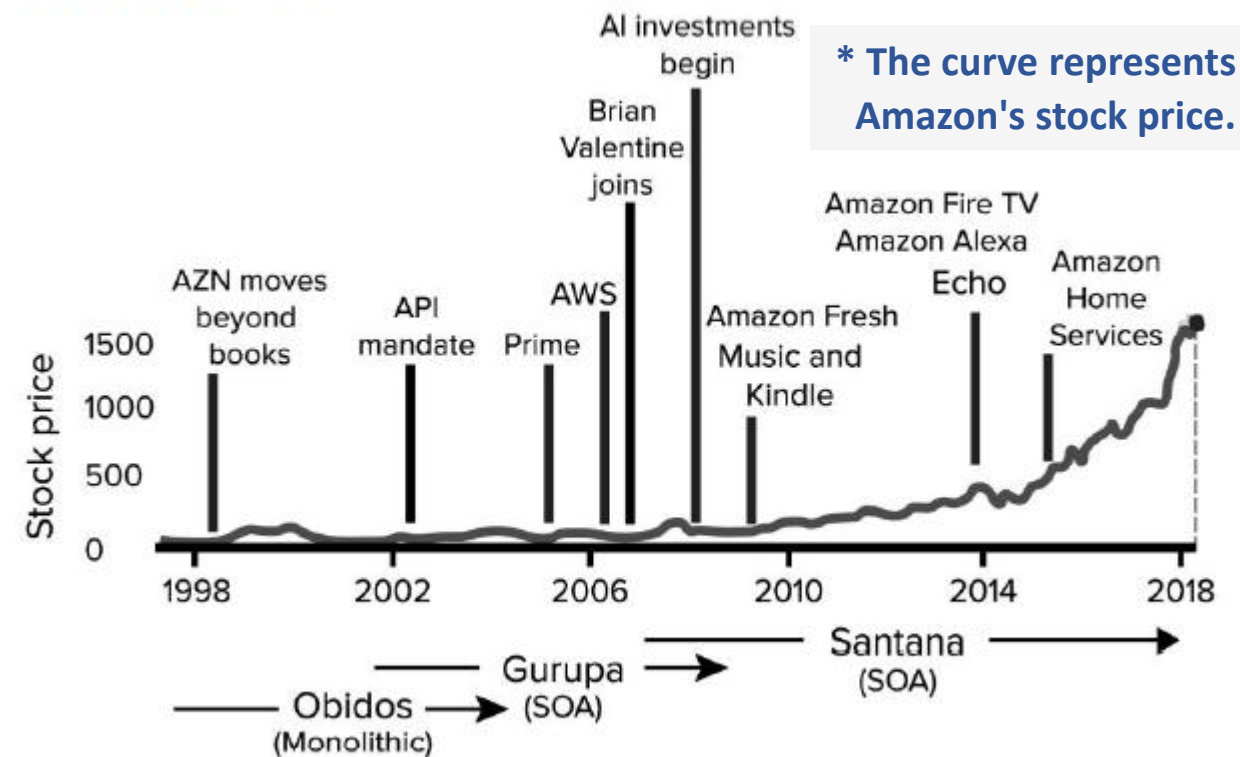
Department-Based Agile Collaboration

Members remain in their home departments and join agile teams temporarily as needed. **They still contribute to original duties.** This often leads to **lack of focus**, poor continuity, communication gaps, misunderstandings, and delayed decisions.

- ❏ **Cross-functional dedicated teams tend to outperform temporary department-based arrangements** in agility, communication, and sustained collaboration.

Building the Platform: Obidos, Gurupa & Santana

FIGURE 4-3



An Amazon time line

The Core Systems

Obidos, Gurupa, and Santana were **systems Amazon built to strengthen its operations and achieve goals** of scale, scope, and learning.

Though **implementation took longer than expected and required Amazon to rewrite nearly all of its e-commerce services**, the effort proved to be well worth it in hindsight, and it propelled them to a **leadership position.**

- Key insight: A real software platform with **centralized, standardized services and clear APIs** is what separates a **scalable enterprise from a collection of disconnected systems.**

The Ideal Digital Operating Foundation

Organizations should be structured to fully leverage digital technology — not just use it as a tool. The ideal state is a shared foundation built **on data, algorithm, APIs, etc.**

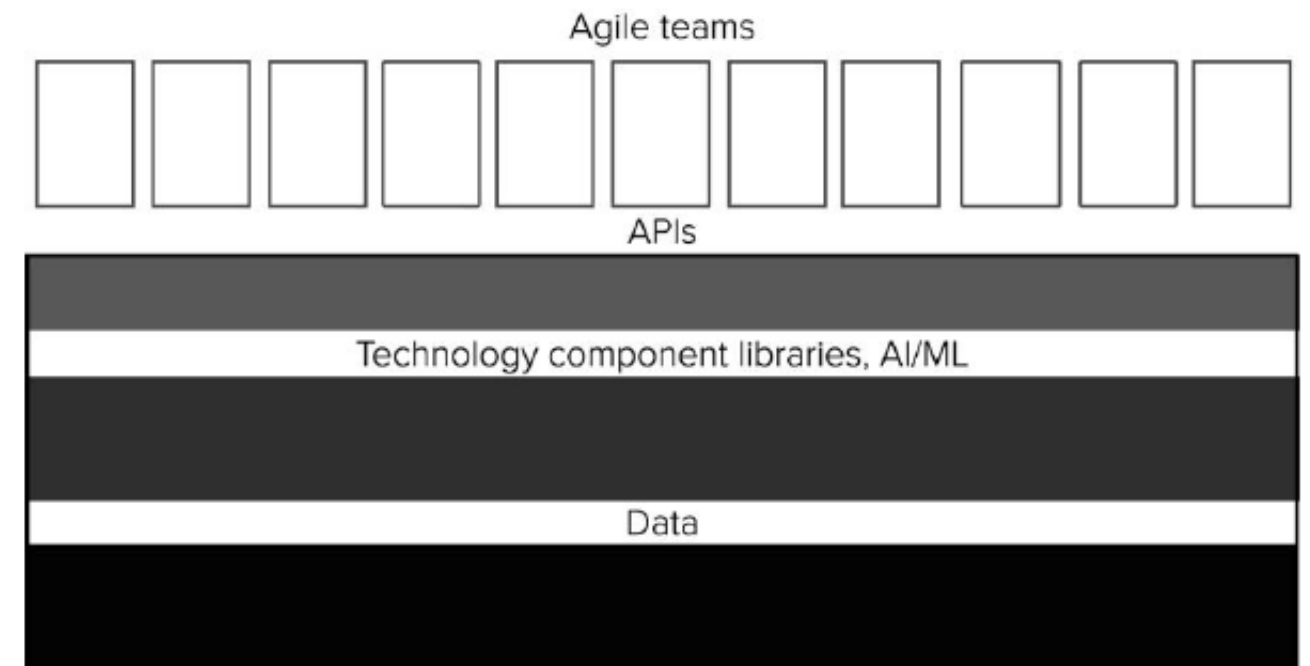
APIs — The Bridge

APIs act **like bridges between different software systems, letting them talk to each other.** A website asking a weather service for data is a simple example — APIs make cross-system collaboration seamless.

Libraries — The Shortcut

Libraries **are pre-written code that developers can import directly, like referencing a formula in a textbook.** They accelerate development by **eliminating repetitive work, especially for AI/ML components.**

FIGURE 4-4



Operating architecture for an AI-powered firm

Two-Pizza Teams on a Shared Foundation

As Amazon's retail **platform was rebuilt**, its development organization evolved into a **modular, distributed structure**. Bezos introduced the **"two-pizza rule"** — **no agile team should be large enough that two pizzas can't feed the whole group (around 5-10 people)** — to eliminate pointless meetings and keep teams nimble.

Independent Operation

Each team works autonomously with its own scope and responsibilities, free to move fast without waiting for others.

Shared Architecture Rules

All teams **follow unified architectural standards, ensuring technical compatibility** and smooth cross-team collaboration.

Code & Data Sharing

Common standards allow teams to **reuse each other's code and share data — critical for powering machine learning and AI at scale.**

The result: **small-team agility *plus* the power of aggregated data** — a combination that fuels Amazon's AI and ML capabilities across the entire organization.

An Entire-New Operating Model

Amazon's transformation **was among the first of a broader economic shift**. From Ant Financial to Google, a generation of AI-driven companies is now designed around this model: aggregating software, data, and analytics to drive scale, scope, and learning.



Zero Marginal Cost Communication

Unlike humans, digital systems ("agents") can communicate with nearly unlimited other agents worldwide at zero marginal cost.



Composable Capabilities

With well-designed shared interfaces, digital systems can connect and combine capabilities — like iOS and Android linking millions of apps and services.



Democratization of Information Services

This model makes high-end tech — computing, storage, databases — **accessible to more people and organizations at low cost, not just large enterprises or tech experts.**

These operating models **represent a radical departure from centuries of business evolution** and pose an existential threat to traditional enterprises.

Breaking Traditional Constraints

In a digital operating model, employees **design and oversee software-automated, algorithm-driven systems that perform the actual delivery** — removing traditional bottlenecks that limit scale, scope, and learning.

The Old Model

Managers supervise employees performing routine tasks.

Growth is constrained by human capacity and organizational boundaries.

The New Model

Managers design and govern digital systems & processes.

Software and algorithms handle execution. Growth is no longer bottlenecked by traditional operations.

The New Role of Managers

AI-driven operating models **fundamentally change the responsibilities of managers.**



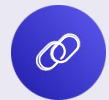
Designers

Shape, refine, and control digital systems that can sense customer needs and respond with value.



Innovators

Imagine how digital systems evolve over time to remain competitive and relevant.



Integrators

Connect independent digital systems and identify new links between business models and their customer service.



Guardians

Maintain the quality, reliability, security, and accountability of the digital systems under their control.

Why Traditional Companies Hesitate

Many traditional companies **remain reluctant to act—**protecting decades of established capabilities, routines, and organizational boundaries.

The Real Obstacles

- They either **don't see their own structural problems**
- Or **are unwilling to commit to the necessary organizational transformation**
- Technology is the easy part—organizational change is the real challenge

- ❑ AI-centric operating models challenge almost all traditional management and operational assumptions—**forcing a fundamental rethinking of what a company is,** how it grows, and the limits of its influence and power.